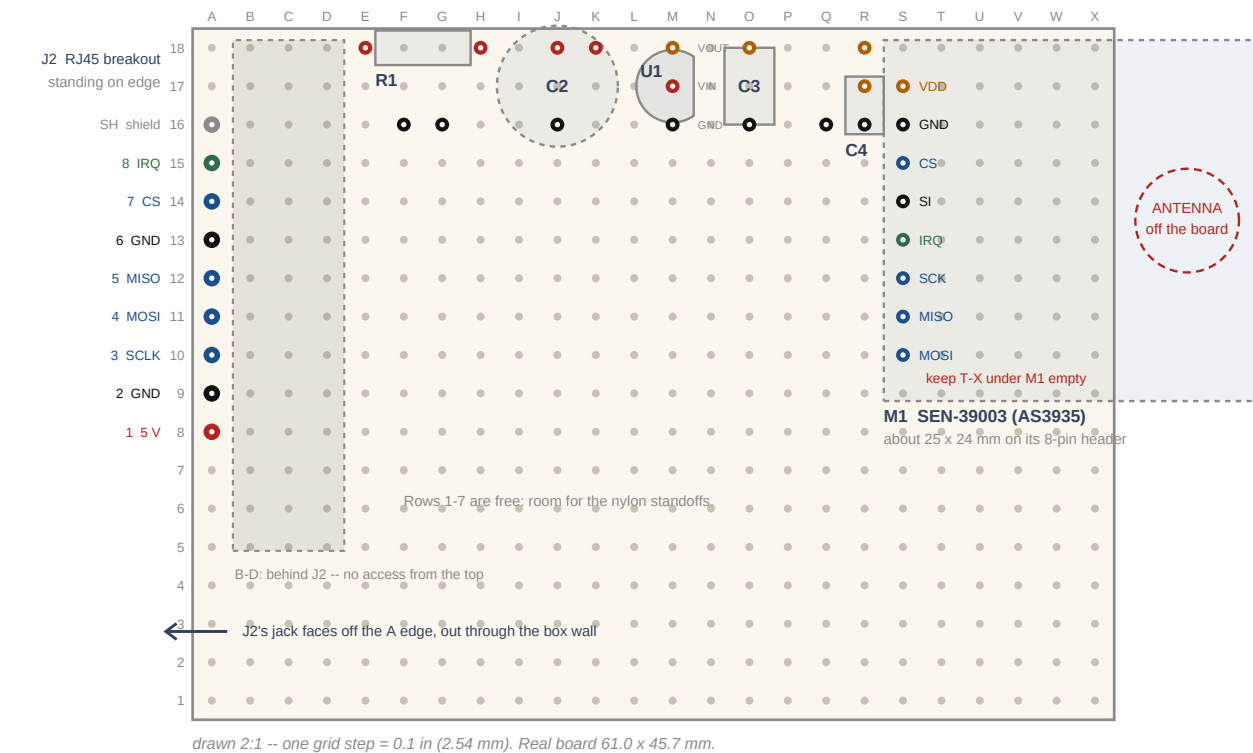


1. SENSOR BOARD -- component placement

1-18 x A-X perf board, A1 at the bottom left, seen from the component side. SELV only: 5 V and SPI.



5 V 3.3 V GND SPI IRQ no connection

Ref	Part	Holes
J2	RJ45 breakout, 9-way header	A8 .. A16, SH at A16, jack off the A edge
R1	100 ohm 1/4 W metal film	E18 - H18
C2	47 uF 50 V, EEU-FR1H470	+ J18 - J16
U1	MCP1700-3302E, TO-92	GND M16 VIN M17 VOUT M18
C3	1 uF X7R, C330C105K5R5TA	+ O18 - O16
C4	100 nF X7R, C320C104K5R5TA	+ R17 - R16
M1	SEN-39003 on an 8-pin header	S10 .. S17, soldered direct

The antenna overhangs the X edge, on purpose

With M1's header in column S its loop antenna sits past the X edge, clear of every pad on the board. Keep T-X under M1 empty, and keep the box wall and its screws clear of it.

C4 sits one hole from VDD and one from GND

That tiny loop is the entire point of the part; do not move it to make room. It is right at M1's edge: keep it low.

U1, TO-92: GND, VIN, VOUT left to right, flat face toward

you and leads down -- not the 78xx order. It stands in column M, flat face toward C3: GND M16 (on PG), VIN M17, VOUT M18 (on BUS-B). L18 stays empty: the isolation gap.

J2 stands on edge in column A

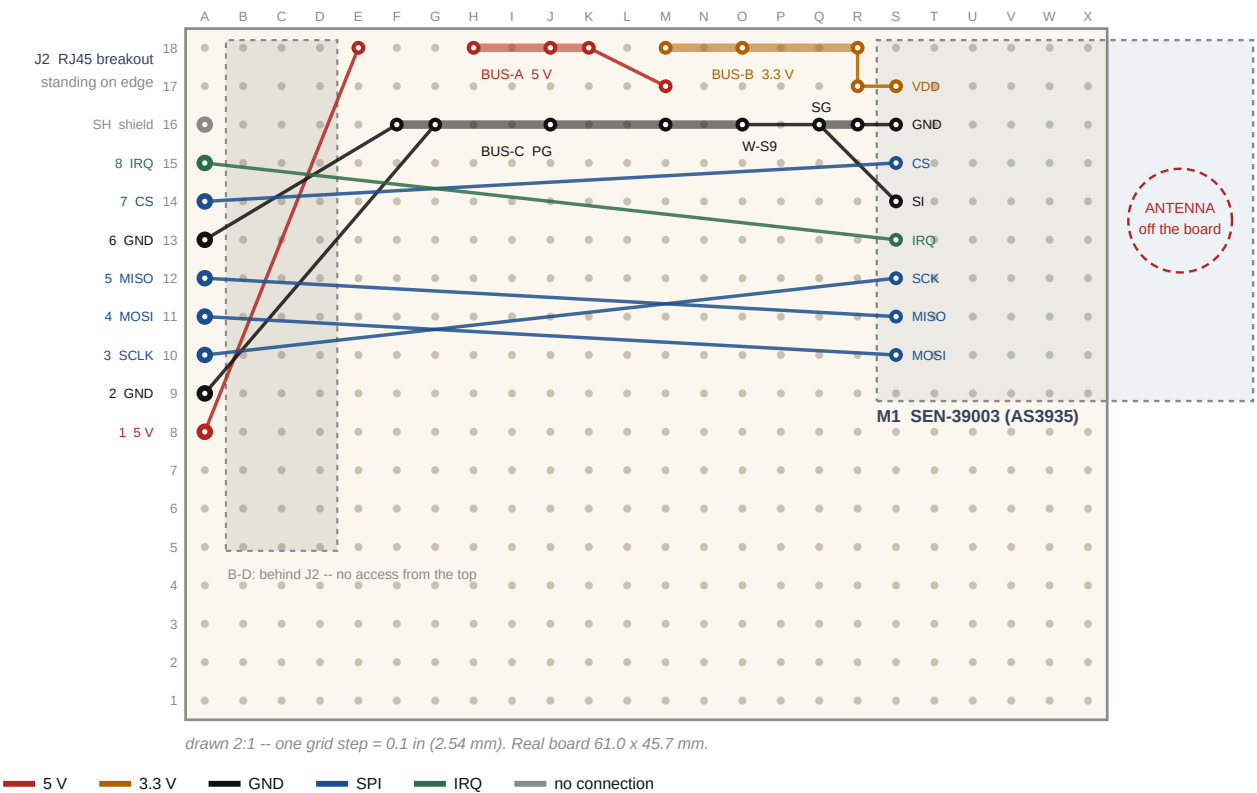
Its right-angle header goes straight into A8-A16, the jack facing off the A edge. Header down, the pins read SH at the top to pin 1 at the bottom. B-D behind it are out of reach from the top, so nothing is placed there -- wires cross them on the back. The wall, not the header, takes plug force.

M1 header order, confirmed on the part

Antenna right, header left, top to bottom: VDD, GND, CS, SI, IRQ, SCK, MISO, MOSI. Soldered straight in, no socket: solderless contacts here are the README 11.3 suspect.

2. SENSOR BOARD -- point-to-point wiring

X-ray view from the component side. All of this is on the BACK of the board, so it mirrors left-right when you flip it over.



Ref	Net	From	To	Wire / note
BUS-A	5 V filtered	row 18	H-K	bare 22 AWG laid across the back of the pads
BUS-B	3.3 V	row 18	M-R	bare 22 AWG laid across the back of the pads
BUS-C	PG power ground	row 16	F-O	bare 22 AWG laid across the back of the pads
BUS-D	SG sensor ground	row 16	Q-R	bare 22 AWG laid across the back of the pads
W-S1	5 V in	A8	E18	J2 pin 1 into R1
W-S2	GND pin 2	A9	G16	onto PG
W-S3	GND pin 6	A13	F16	onto PG -- the SCLK return
W-S4	LDO VIN	K18	M17	BUS-A into U1 VIN, past L17-L18
W-S5	3.3 V	R18	R17	BUS-B down to the C4 / VDD node
W-S6	VDD link	R17	S17	one hole -- do not lengthen
W-S7	GND link	S16	R16	sensor GND onto SG
W-S8	SI strap	S14	Q16	grounds SI: selects SPI, not I2C
W-S9	SG-PG tie	O16	Q16	the ONLY tie -- over P16
W-S10	SCLK	A10	S12	
W-S11	MOSI	A11	S10	
W-S12	MISO	A12	S11	
W-S13	CS	A14	S15	
W-S14	IRQ	A15	S13	

Two grounds, one tie

PG carries the cable's ground return, the bulk cap and the LDO reference. SG carries only the sensor's GND pin, its 100 nF and the SI strap. They meet at exactly one place: W-S9, over the empty P16.

Bridge them anywhere else and you have wrapped a ground loop around the LDO; the 100 nF stops being local and the whole point of the split is gone.

C3 references PG, C4 references SG

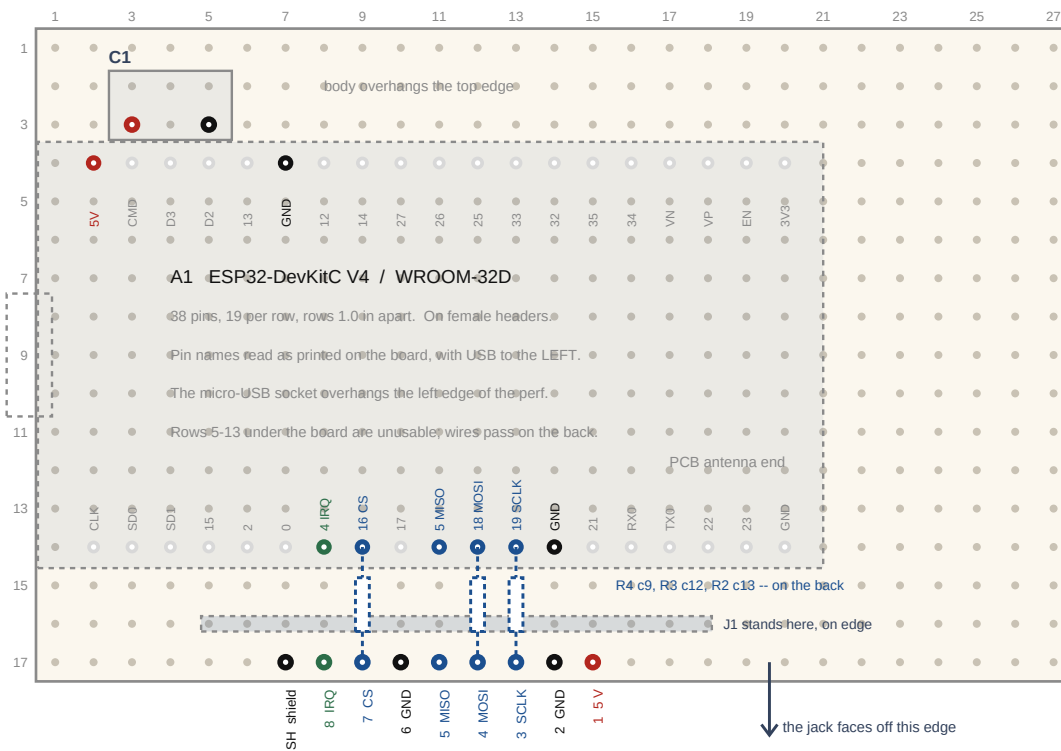
C3 is the MCP1700's stability capacitor, so it belongs to the regulator and lands on PG. C4 is the sensor's decoupling, so it belongs to the sensor and lands on SG.

Crossings are fine, except over a bus

Point-to-point links are insulated and run on the solder side; they cross each other freely -- the SPI links all run in rows 10-15, below every bus. The four buses are BARE.

3. MAIN BOARD -- component placement

Contents: the ESP32 dev board, the 5 V bulk cap at its pins, three series terminators, and J1, the RJ45 breakout.



drawn 2:1 -- one grid step = 0.1 in (2.54 mm). Real board 68.6 x 43.2 mm.

5 V 3.3 V GND SPI IRQ no connection

Ref	Part	Holes
A1	ESP32-DevKitC V4 (WROOM-32D)	female headers: row 4 c2-20, row 14 c2-20
C1	470-1000 uF 16-25 V 105 C	+ (3,3) - (5,3) stripe at (5,3)
J1	RJ45 breakout, 9-way header	row 17 c7-15, SH at c7, jack off the edge
R2	SCLK series, 68 ohm 1/4 W	(13,14) - (13,17), pin to pin, back
R3	MOSI series, 68 ohm 1/4 W	(12,14) - (12,17), pin to pin, back
R4	CS series, 68 ohm 1/4 W	(9,14) - (9,17), pin to pin, back

The jack faces off the bottom edge -- that fixes the order

J1's right-angle header goes straight into row 17, so the breakout stands on edge. Header down, looking into the jack, the pins read SH 8 7 6 5 4 3 2 1 -- so SH sits at col 7, the USB end. Before soldering, check the jack overhangs the edge.

The header is the electrical joint, not the mechanical one

Nine pins in isolated pads will lift under a cable being plugged in. The enclosure wall, or a bracket on J1's own mounting holes, must take that force. J1 stands about 30 mm tall.

Socket the ESP32, unlike the sensor. Check its row spacing (drawn at 10 holes, 1.0 in) and use it as its own jig.

These are not the default SPI pins, on purpose

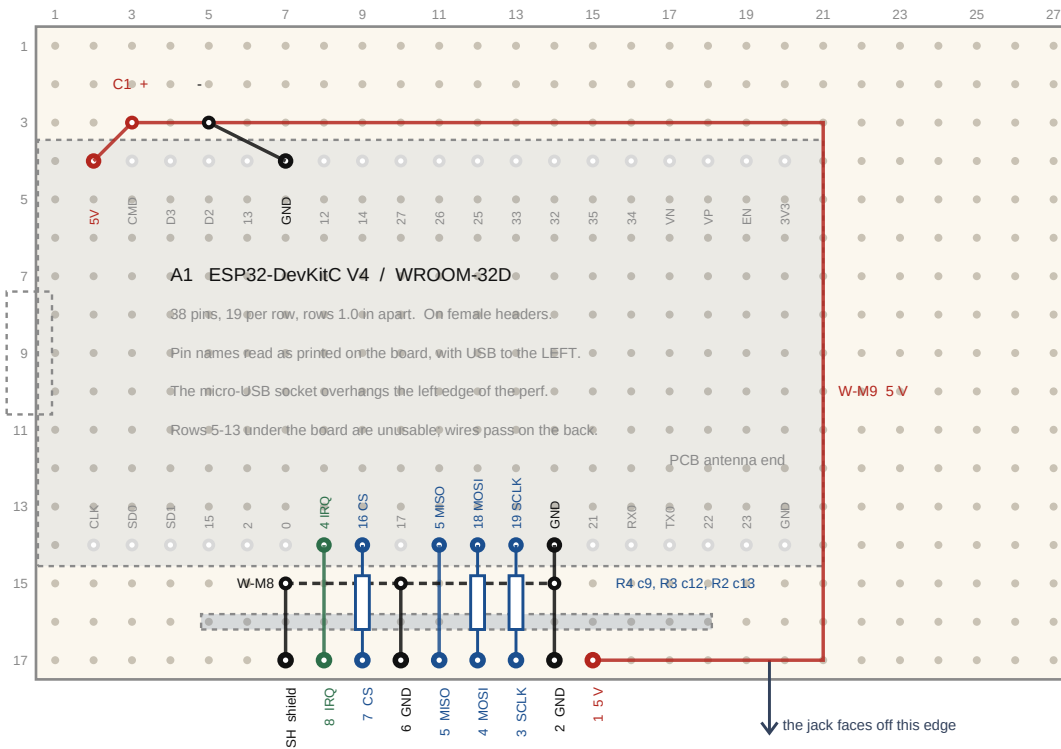
Each resistor sits pin to pin between an ESP32 pin and the jack pin directly below it, and neither pin order can move. So the GPIOs were chosen to line up with the jack: SCLK 19, MOSI 18, MISO 5, CS 16, IRQ 4. The ESP32 routes SPI to any pin; at these speeds the GPIO matrix costs nothing. lightning-detector.yaml matches -- do not 'fix' it.

C1 is as tight as the DevKitC allows

Five pins between 5V and its nearest ground: ~27 mm of loop however you arrange it. Keep W-M1 and W-M2 short.

4. MAIN BOARD -- point-to-point wiring

X-ray view from the component side. Solid lines are on the BACK; the one dashed line, W-M8, is on the component side.



drawn 2:1 -- one grid step = 0.1 in (2.54 mm). Real board 68.6 x 43.2 mm.

5 V 3.3 V GND SPI IRQ no connection

Ref	Net	From	To	Note
W-M1	C1+ to 5V	(3,3)	(2,4)	as short as it will go
W-M2	C1- to GND	(5,3)	(7,4)	as short as it will go
W-M3	IRQ	(8,14)	(8,17)	GPIO4, pin to pin, no resistor
W-M4	MISO	(11,14)	(11,17)	GPIO5, pin to pin, no resistor
W-M5	GND, pin 2	(14,14)	(14,17)	solder to (14,15) on the way
W-M6	GND, pin 6	(10,17)	(10,15)	stub to the W-M8 tap
W-M7	shield	(7,17)	(7,15)	stub to the W-M8 tap
W-M8	ground hop	(7,15)	(14,15)	COMPONENT SIDE; tap at (10,15)
W-M9	5 V to cable	(3,3)	(15,17)	via (21,3), (21,17): round the header
R2	SCLK, 68 ohm	(13,14)	(13,17)	pin to pin on the back
R3	MOSI, 68 ohm	(12,14)	(12,17)	pin to pin on the back
R4	CS, 68 ohm	(9,14)	(9,17)	pin to pin on the back

Everything between the headers is on the back

Rows 15-16 have no room for a footprint, so the three resistors and three straight links solder pin to pin, lying flat on the back between the ESP32 joint and the jack joint below. Cols 11-14 are four parallel runs 2.54 mm apart: insulated wire for the links, short straight leads on the resistors.

Why one wire is on the component side

Those six runs wall off the back of rows 15-16, so pin 6 (col 10) and SH (col 7) cannot reach ground at col 14 without crossing them. W-M8 hops over the top instead. Fit it BEFORE J1: afterwards the breakout stands over row 16 and it is buried.

5 V goes round the end of the header

W-M9 leaves C1 along row 3, down col 21 and back along row 17. Every other way to row 17 passes between two header pins whose solder joints are a millimetre apart. It is long; at ~1 mA that costs nothing.

Three cable grounds, one ESP32 pin

Pins 2, 6 and SH all meet at the GND beside GPIO19 -- the pin next to SCLK, so the SCLK pair (3,6) returns where it should. C1 keeps the top-row GND to itself. SH is bonded here and nowhere else; at the sensor board it stays floating.

5. Build order, and the things that will bite

Read this before you cut anything. Sections refer to the project README.

Sensor board

1. The 1-18 x A-X board, A1 at the bottom left as printed. Every coordinate is letter then number, as on the board.
2. Buses first, while the board is flat and empty: PG (row 16, F-O), SG (row 16, Q-R), 5 V (row 18, H-L), 3.3 V (row 18, N-R).
3. R1, C2, U1 (flat face toward C3), C3, C4. Then W-S9, the single SG-PG tie: do it deliberately and mark it.
4. Solder the 8-pin header into the SEN-39003, then that assembly into S10..S17, antenna out past the X edge.
5. J2: header into A8..A16, SH at the top, jack off the A edge. Nothing goes in B-D behind it.
6. On the back: W-S8, the SI strap -- without it the part talks I2C and nothing works -- then the power and signal links.
7. Before power: ohmmeter 3.3 V to either ground. Under a few hundred ohms means a bridge -- find it now, not later.

Main board

1. Cut the perf to 27 x 17 and mark hole (1,1) with a pen: this board is counted, not lettered. Measure the dev board's row spacing, then solder the female headers using it as the jig.
2. C1, stripe at (5,3). Then W-M1 and W-M2, short.
3. W-M8 on the COMPONENT side, now, while row 15 is reachable: insulated, stripped at (7,15), (10,15) and (14,15).
4. J1: SH at col 7, jack overhanging the bottom edge. Check the orientation twice, then solder its header into row 17.
5. On the back: R2/R3/R4 and W-M3/4/5 pin to pin, then the W-M6 and W-M7 stubs up to the W-M8 taps.
6. W-M9, the 5 V run, round the end of the header via col 21.
7. Power it with NO ESP32 in the socket: 5 V at J1 pin 1, and nowhere it should not be.

Things that will bite

The X-ray thing. Both wiring pages are drawn as if you could see through the board from the top, because that is how you place parts. Flip the board to solder and left/right swap. This is the single most common perf-board error.

Place by the printed label. The sensor board reads A1 at the bottom left and is drawn that way up, but the label printed beside each hole is still the thing to trust.

Isolated pads, not strips. If what you have is stripboard, every row here is already a bus and the layout is wrong without track cuts. Check before you buy.

RJ45 on 0.1 in perf. A bare RJ45 jack does not fit; the breakout's 9-way header does, on both boards, standing on edge with the jack off the edge. The 3 rows or columns behind it are then out of reach from the top: fit anything there first. Wire to the printed numbers, never to a position in a photo.

This jack is not Ethernet. It carries 5 V and SPI. A live PoE port puts 48 V on those lines and destroys both ends. Label both boxes now, while you are holding them -- README 7.1.

Nylon standoffs, nylon screws on the sensor board especially. A steel screw beside a 500 kHz loop antenna is a shorted turn.

Solid wire, not the silicone stranded. Buses have to be bare solid or they are not straight; links have to be solid or they will not enter a hole without tinning. README 7.6.

Rigidity is a measurement, not a feeling. README 11.3: the breadboard's noise floor fell by two thirds the moment the build was handled. Phase 2 in README 15 is survey, handle the box, survey again. If the rates move, the build is furniture and nothing measured on it means anything.

Keep data_rate: 200kHz in lightning-detector.yaml. It is set; the ESPHome default is 1 MHz. README 7.1.



2.000 in / 50.8 mm -- 20 holes at 0.1 in pitch. If this does not measure true, your printer scaled the page.